Circuit Documentation

1. Circuit Summary

This circuit is an ESP32-based wireless interface using a CC1101 sub-GHz radio module. The ESP32 communicates with the CC1101 over an SPI bus and uses two additional GPIO pins for the CC1101 data-ready/control signals.
The circuit includes:

- ESP32 WROOM-32U microcontroller
- CC1101 wireless transceiver module
- External antenna connected to the CC1101
- PCB antenna connected to the ESP32 antenna connection
- Orange status LED driven by ESP32 GPIO2
- 220 Ω current-limiting resistor for the LED

The CC1101 is powered from the ESP32 3.3 V supply and shares ground with the ESP32. The circuit contains no separately documented firmware code.
Design note: The LED is connected from GPIO2 through a 220 Ω resistor to ground. Depending on the LED characteristics and ESP32 output voltage, the GPIO output current should be checked to ensure it remains within the ESP32 GPIO limits.

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2. Component List

ESP32 WROOM-32U

The ESP32 is the main microcontroller in the circuit. It provides:

- 3.3 V power to the CC1101
- Ground reference
- SPI communication with the CC1101
- Control and interrupt connections to the CC1101
- GPIO2 control for the orange indicator LED
- Connection to a PCB antenna through its antenna pin

No firmware code was supplied for this microcontroller.

CC1101 Module

The CC1101 is a sub-GHz wireless transceiver connected to the ESP32 through SPI.
Its connections include:

- 3.3 V power
- Ground
- SPI clock
- SPI data input and output
- Chip-select control

- Two general-purpose digital I/O signals
- External antenna connection

HC12 ANTENNA

This is the external antenna connected to the antenna input of the CC1101 module.

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pcb_antenna

This antenna is connected to the ESP32 WROOM-32U antenna connection.

LED: Two Pin (orange)

The orange LED provides a visual status indicator. Its anode is connected to the ESP32 GPIO2 output through a 220 Ω resistor, and its cathode is connected to ground.

Resistor — 220 Ω

The resistor limits current through the orange LED. It is connected in series between ESP32 GPIO2 and the LED anode.

Comment V2 Elements

Several Comment V2 elements are included in the source design. They have no electrical pins and therefore do not participate in circuit operation or wiring.

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3. Component Wiring Details

ESP32 WROOM-32U

ESP32 Pin	Connected To	Function
3v3	CC1101 Vcc	3.3 V power supply for the radio module
GND	CC1101 GND and LED cathode	Common circuit ground
23	CC1101 MOSI	SPI data from ESP32 to CC1101
18	CC1101 SCLK	SPI clock
19	CC1101 MISO	SPI data from CC1101 to

ESP32 Pin	Connected To	Function
		ESP32
27	CC1101 GDO2	CC1101 general-purpose digital output/control signal
4	CC1101 GDO0	CC1101 general-purpose digital output/control signal
15	CC1101 CSN	SPI chip-select, active-low
2	Resistor pin 2	LED drive output
Antenna	pcb_antenna wire	ESP32 antenna connection

The following ESP32 pins are not connected in the supplied net list:

- EN
- VP
- VN
- 34
- 35
- 32
- 33
- 25
- 26
- 14
- 12
- 13
- D2
- D3
- CMD
- 5V
- 22
- TX
- RX
- 21
- 17
- 16
- 0
- 15 is connected to CC1101 CSN
- D1
- D0
- CLK

The ESP32 GND pin is connected to both the CC1101 ground and the LED cathode. The net list does not distinguish between multiple physical ground pins on the ESP32 module.

CC1101 Module

CC1101 Pin	Connected To	Function
Vcc	ESP32 3v3	3.3 V power

CC1101 Pin	Connected To	Function
GND	ESP32 GND	Ground reference
ANT	HC12 antenna ANT	RF antenna connection
MOSI	ESP32 23	SPI data input
SCLK	ESP32 18	SPI clock input
MISO	ESP32 19	SPI data output
GDO2	ESP32 27	General-purpose digital I/O
GDO0	ESP32 4	General-purpose digital I/O
CSN	ESP32 15	SPI chip-select input

The CC1101 pin labeled Gnd is not connected in the supplied net list. Only the pin labeled GND is connected to the ESP32 ground.

HC12 ANTENNA

Antenna Pin	Connected To	Function
ANT	CC1101 ANT	RF connection to the CC1101 module

The antenna is connected directly to the CC1101 antenna pin as represented in the supplied net list.

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pcb_antenna

Antenna Pin	Connected To	Function
wire	ESP32 Antenna	RF antenna connection for the ESP32

LED: Two Pin (orange)

LED Pin	Connected To	Function
anode	Resistor pin 1	LED positive terminal
cathode	ESP32 GND	LED return path

The LED illuminates when ESP32 GPIO2 is driven high, subject to the LED polarity and the GPIO output current.

Resistor — 220 Ω

Resistor Pin	Connected To	Function
pin1	LED anode	Series connection to the LED
pin2	ESP32 2	GPIO2-driven LED input

The resistor provides current limiting for the orange LED.

Comment V2 Elements

The comment elements have no pins and no electrical connections. They are documentation-only objects in the source design.

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4. Documented Code

No microcontroller code was supplied with the circuit input.
The ESP32 firmware would need to configure and use the following interfaces:

- SPI:
  - MOSI: GPIO23
  - MISO: GPIO19
  - SCLK: GPIO18
  - CSN: GPIO15
- CC1101 digital signals:
  - GDO2: GPIO27
  - GDO0: GPIO4
- Status LED:
  - GPIO2

A complete software implementation would typically initialize the SPI peripheral, configure the CC1101 registers and radio frequency parameters, handle the GDO signals, and control the LED as a status indicator.